Assignment -1
Wine Quality Classification Using Machine Learning

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I. OBJECTIVE

The primary goal of this assignment is to:
1. Train and test two classifiers Random Forest and Gradient Boosting) on the Wine Quality Dataset to predict wine quality.
2. Handle class imbalance by using SMOTE for oversampling the minority classes.
3. Compare the performance of the classifiers using F1-macro as the evaluation metric.
4. Choose the best-performing classifier and evaluate its performance on the test set.

II. DATASET ANALYSIS

A. Dataset Overview

- The data set includes information on wine's physico-chemical properties and quality scores (from 3 to 9).
- The target variable quality is imbalanced, as shown by the initial class distribution.

B. Class Distribution Before Balancing

- Quality 6 is the most common, followed by quality 5.
- Minority classes include quality 3 and 9, with very few samples.

Training Class Distribution:
6: 1758, 5: 1166, 7: 704, 8: 140, 4: 130, 3: 16, 9: 4

III. CHOICE OF METRIC

The instructions require selecting the best metric to evaluate the task of wine quality classification. After careful consideration, F1-macro was chosen as the primary evaluation metric for the following reasons:

A. Handling Imbalanced Classes

- The dataset shows a considerable class imbalance, with quality 6 and 5 dominating and 3 and 9 being rare. Accuracy alone may not accurately reflect performance in this scenario, as it could be achieved by correctly predicting majority classes while ignoring minority classes.

B. Balance Between Precision and Recall

- The F1-score balances Precision (accurately predicted positives) and Recall (number of actual positives recorded).
- Accurately collecting these aspects is critical for wine quality classification, as missing or incorrectly predicting specific quality classes can have practical implications.

C. Macro Averaging

- F1-macro calculates and averages F1-scores for each class, ensuring equal treatment for all. This approach prioritizes minority classes alongside majority classes, aligning with the task's objectives.

IV. METHODOLOGY

1) To retain class distribution, the dataset was divided into 80% training and 20% test sets.
2) To standardize the data, the features were scaled with StandardScaler.
3) Cross-validation (CV):
   - Cross-validation was performed using Repeated Stratified K-Fold CV with n_splits=3 and n_repeats =10 to ensure robust evaluation.
   - Evaluated both models Random Forest and Gradient Boosting) using F1-macro.
4) Handling Class Imbalance:
   - Applied SMOTE to oversample minority classes in the training set.
   - Ensured balanced class distribution after SMOTE.

   Balanced Class Distribution:
   6: 1758, 5: 1166, 7: 704, 8: 200, 4: 200, 3: 100, 9: 100

V. Results

A. Performance on Imbalanced Data

Model	Mean F1 (Macro)	Standard Deviation
Random Forest	0.3604	0.0164
Gradient Boosting	0.3261	0.0191

TABLE I
Model Performance Metrics

- Both models performed poorly on the imbalanced dataset due to the skewed class distribution. Minority classes were not well-represented, leading to lower F1-macro scores.

B. Performance on Balanced Data (After SMOTE)

Model	Mean F1 (Macro)	Standard Deviation
Random Forest	0.6949	0.0143
Gradient Boosting	0.5705	0.0167

TABLE II
Model Performance Metrics

- After balancing, both models improved significantly in F1-macro scores.
- Random Forest outperformed Gradient Boosting, resulting in a higher mean F1-macro score and decreased variability.

C. Test Set Evaluation (Final Model - Random Forest)

Quality	Precision	Recall	F1-Score	Support
3	0.00	0.00	0.00	4
4	0.62	0.30	0.41	33
5	0.69	0.64	0.66	291
6	0.63	0.79	0.70	440
7	0.70	0.49	0.58	176
8	0.95	0.57	0.71	35
9	0.00	0.00	0.00	1
Overall	0.67	0.66	0.66	980

TABLE III
Classification Metrics by Class

- Random Forest achieved an overall weighted F1 score of 0.66 on the test set.
- The model performed well for the majority of classes (e.g., quality 6, 5) but struggled with minority classes (3, 9), which is expected due to their very low representation.

VI. Comparison Summary

Metric	Random Forest	Gradient Boosting
Imbalanced F1 (Macro)	0.3604	0.3261
Balanced F1 (Macro)	0.6949	0.5705
Final Model (Test F1)	0.66(weighted)	N/A

TABLE IV
Metrics Comparison Between Random Forest and Gradient Boosting

- Balancing the data with SMOTE significantly improved both models' performance.
- Random Forest outperformed Gradient Boosting for both balanced and imbalanced conditions.

VII. Findings and Discussion

A. Effect of Balancing

- SMOTE successfully addressed class imbalance, leading to improved performance in minority classes.
- Random Forest benefited more from balancing compared to Gradient Boosting.

B. Random Forest vs. Gradient Boosting

- Random Forest emerged as the better-performing model, both in cross-validation and final test evaluation.
- Its ensemble nature, with a focus on bagging and randomness, might have provided robustness to the dataset's variability.

C. Challenges with Minority Classes

- Despite SMOTE, minority classes (3 and 9) are challenging to predict due to their low occurrence in the test set.
- Further techniques, such as cost-sensitive learning, could help improve predictions for such rare classes.

D. F1 Macro as Metric

- The choice of F1 macro was effective in evaluating imbalanced data, as it ensured fair evaluation across all classes.

VIII. Conclusion

- **Best Model:** Random Forest
- **Final Performance:** Weighted F1-Score of 0.66 on the test set.
- Balancing the data was crucial for achieving reasonable performance, highlighting the importance of addressing the class imbalance in real-world datasets